

Glossary

V. 5.3

See also (Android) <https://androidaps.readthedocs.io/en/latest/UsefulLinks/Glossary.html#glossary> and (iOS) <https://loopkit.github.io/loopdocs/faqs/glossary/>



For an introduction into the topic of **Artificial Pancreas Systems** (“Looping”), see:

- <https://github.com/danamlewis/artificialpancreasbook/>
- and <https://androidaps.readthedocs.io/en/latest/Resources/clinician-guide-to-AndroidAPS.html#for-clinicians-a-general-introduction-and-guide-to-aaps>.
- Overview over all DIY loops <https://www.diabettech.com/user-resources/hcp-loop-guide/>
- For a resource on key topics like ISF, meal management etc. see the pdf collection in the HCL branch of: <https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings>
- The parallel (default) branch with a “**FCL-e-book**” <https://github.com/bernie4375/FCL-potential-autoISF-research-> is for advanced users of systems with SMB+UAM (oref) algorithm.

Term	Description	related	more details @
AAPS	AAPS is the name of an Open Source (aka “DIY”) looping app: On Android phones, Bluetooth connected with an insulin pump and a CGM , it provides an Artificial Pancreas System Broadest choice of pumps and CGMs of any looping option Algorithm: see oref , UAM	oref iAPS, Trio iOS Loop	https://androidaps.readthedocs.io/en/latest/introduction.html#what-is-android-aps-aaps
AAPS Client	AAPS <u>can be monitored and controlled remotely</u> via the AAPSClient app and optionally via the associated Wear app running on Android Wear watches <i>Similar for iOS Loop -> Loop caregiver</i>	<i>iOS equivalent:</i> Loop Caregiver app	https://androidaps.readthedocs.io/en/latest/GettingStarted/FAQ.html#configuring-and-using-the-aapsclient-remote-app https://androidaps.readthedocs.io/en/latest/RemoteFeatures/RemoteMonitoring.html
acce_ISF	...modulates ISF to boost iob at beginning bg rises; tuneable via bgAccel_ISF_weight <i>iAPS: “ISF weight while bg accelerates”</i>	autoISF	see section 4.2. in: “ FCL-e-book ” https://github.com/bernie4375/FCL-potential-autoISF-research-
acceleration	mathematical analysis of the bg development can reveal earliest signs of a bg rise; this is highly relevant in -> FCL w/ autoISF ; -> parabola fit . Growing bg deltas are a simpler way of detecting acceleration, but with a ~ 10-20 minutes relative delay. A decelerating rise (negative acceleration value) indicates a bg peak will soon be reached.	brake	https://github.com/gazelle/APS-what-if/blob/A3.2.0.4_ai3.0.1_Quick_Guide.pdf FCL-e-book , section 4.2

Activity Monitor	feature of some loop systems that allow adaptation of loop aggressiveness with ~ past hour data from the phone's (or watch') motion monitor (evtl also heart rate).	aggressiveness	FCL-e-book , section 5.1.5
aggressiveness of the loop	more aggressive loop settings will deliver more insulin, often via a lowered temporary ISF being applied to a needed correction, or also via a temp. lowered bg target . A more aggressive loop helps fight temp. insulin resistance (e.g. after fatty meals). Conversely, e.g. in an exercise context, higher ISF and higher temp.glucose target help deal with increased insulin sensitivity , and deliver less insulin.	sensitivity adaptation resistance temp.profile override	https://androidaps.readthedocs.io/en/latest/DailyLifeWithAaps/ProfileSwitch-ProfilePercentage.html https://loopkit.github.io/loopdocs/operation/features/overrides/ FCL-e-book , section 5
AIMI	dev variant of AAPS and of Trio involving simple Meal Announcement (MA) that might be stretched into a FCL		FCL-e-book 13.3.2 https://discord.gg/TPDQzS3Bq3
algorithm	the algorithm is a set of calculations and plausibility/safety checks the loop goes through every 5 minutes (upon receipt of a new CGM value), to define what to do, notably in terms of more insulin delivery for control of bg (to bring it to target). DIY looping algorithms see: • -> iOS Loop; • -> oref (OpenAPS origin)	control; oref; iOS Loop insulin kinetics	https://www.diabettech.com/looping-a-guide/comparing-the-loop-and-openaps-algorithms/
AMA	advanced meal assist - algorithm to handle carbs via % TBR (loop not giving small boli) <i>iOS Loop equivalent</i> : "Temp.Basal Only Dosing Strategy"	TBR	Wiki - AMA https://loopkit.github.io/loopdocs/operation/algorithm/auto-adjust/#determine-the-temporary-basal-rate
Android Studio	(free) developer software needed to complete and maintain your personal copy of AAPS	Github <i>for i-Phone loops:</i> Xcode	https://androidaps.readthedocs.io/en/latest/Installing-AndroidAPS/troubleshooting-androidstudio.html#troubleshooting-android-studio
Anubis	DIY re-engineered transmitter for Dexcom G6 CGM ; lasts unlimited (w/ battery change ~ every 6 months); will not shut down sensor at 10.0 days (as factory transmitters do). For more info: „Followers of Anubis“ Facebook group	G6 G6 x 2	https://docs.google.com/forms/d/e/1FAIpQLSdGtAmwqkBUaMVbBPENF_eRBSz7ZMcCz-3CjLxwc4TC6_RH5w/viewform
apk	software installation file (Android application package)	Github	Wiki - Building APK
APS	Artificial Pancreas System. Semi-automatic insulin delivery system that, coupled with a CGM , can regulate bg to target. Besides DIY systems (OpenAPS,	AAPS; Trio; iAPS;	https://iaps.readthedocs.io/en/latest/resources/alternative.html#comparison-table-

	iOS Loop, AAPS, iAPS and Trio) that pioneered this area, there is an increasing number of commercial systems now available	iOS Loop CGM	of-automated-insulin-delivery-systems ; https://github.com/danamlawis/artificialpancreasbook/
Artificial Pancreas System (APS)	a system which works to automatically keep blood sugar levels within healthy limits: by detecting glucose levels , using these values to do calculations , and then delivering the (predicted) right amount of insulin to the body. It repeats the calculation, every few minutes, 24/7.		https://androidaps.readthedocs.io/en/latest/introduction.html#what-is-an-artificial-pancreas-system
autoCR	automated carb ratio adjustment (like autoISF or dynamicISF for ISF modulation). However, in oref loops, doing this for carb ratio is less useful. Pilot implementation available in iAPS, see reference		autoCR is discussed in autoISF section of iAPS Discord: https://discord.gg/R4EHVya2cB
autoISF	oref SMB+UAM, with very sharp adaptation of ISF to glucose “behavior” (acceleration , delta , level, stuck-at- high). Ideal for FCL but difficult to set up (initial “tuning”). Useful also in HCL (tuning then different) autoISF is available only in dev variants of 1. AAPS 2. iAPS 3. Trio	FCL	https://github.com/gazelle/autoISF/blob/A3.3.3.a+aisf3.1.0 Quick Guide.pdf or newer 1. https://github.com/T-o-b-i-a-s/AndroidAPS/ 2. https://github.com/mountrcg/iAPS 3. https://github.com/mountrcg/Trio FCL-e-book
Auto(matic) Bolus	small bolus given by the loop: advanced feature for faster bg adjustment than via TBR —only (see AMA); given bolus size is limited to 40% of the calculated bolus (or to the set max.bolus, if smaller). Zero basal right after an automatic bolus is expected	<i>oref</i> = SMB	https://loopkit.github.io/loopdocs/operation/features/carbs/?h=auto+bolus#automatic-bolus https://loopkit.github.io/loopdocs/operation/algorithm/auto-adjust/#deliver-automatic-bolus-with-scheduled-basal
Automation (Feature integrated in AAPS; other loops may need 3 rd party software; or “middleware”)	1. analyze patterns in YOUR data, (at times, geo-locations, or bg and iob patterns that point to a problem ...) where you want your loop act differently: carve out Conditions that describe the situations	Automated aggressive-ness modulation DIY cockpit <i>iOS loops equivalent:</i> middleware	https://androidaps.readthedocs.io/en/latest/Usage/Automation.html#automation https://androidaps.readthedocs.io/en/latest/Usage/automation-withapp.html#automation-with-third-party-android-automate-app

	2. Define Actions (loop settings for different aggressiveness) for x minutes Specifically in AAPS: User Action Automations enable -> DIY cockpit		“Uses ... of Automations.pdf” at: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings
Autosens	calculation of sensitivity to insulin as a result of exercise, hormones etc. in the past 8 – 24 hrs, and automatic % adjustment (within selected min and max borders) of basal, ISF and (if selected) bg targets. ((Note, if Autotune is also selected, the result from Autosens will be used to adjust the profile, rather than temp. moderating key profile parameters))	iob delta Autotune	DIABETTECH - Autosens https://openaps.readthedocs.io/en/latest/docs/Customize-Iterate/autosens.html?highlight=Open-APS%20Autosens#notes-about-autosensitivity
Autotune	Autotune can be used to get suggestions how to tune profile basal; it gives also one 24h average IC and ISF suggestion. Controversial (see 3rd link given)! Not for use with dynamicISF, autoISF.	Autosens	https://androidaps.readthedocs.io/en/latest/Usage/autotune.html#how-to-use-autotune-plugin-dev-only https://iaps.readthedocs.io/en/main/settings/configuration/autotune.html#autotune https://github.com/Artificial-Pancreas/iAPS/pull/1006 “Autotune...pdf” in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings/blob/HCL-.settings-main-repo-(pdf)/
average delta	see delta (long or short avg. delta bg)	.	
basal rate	the basal rate defined in the profile* (that you give to your loop to work with) is the amount of hourly insulin to maintain bg at a stable level, in absence of -> disturbances *note that basal might get auto-adjusted (e.g. by Autosens, or in an exercise mode).	IC / ISF profile disturbance	
bg	blood glucose: the tissue glucose that all CGMs measure reflects the blood glucose, with a couple of minutes of delay. (This, plus the minutes of spacing between CGM values, adds to the “sluggishness” of getting our bg regulated by the loop).	control (sluggishness) Libre 3	
bg_delta	see delta		
bg deviation	describes deviations when the observed changes in glucose do not quite match up with the expected change due to insulin effects alone -> $\text{absorbed_carbs} = \text{bgDEV} * \text{IC} / \text{ISF}$	called insulin counteraction effect in iOS Loop	https://androidaps.readthedocs.io/en/latest/Usage/COB-calculation.htm
bg_ISF	at high bg levels, stronger (lower) ISFs may be useful, and milder ones if $\text{bg} < \text{target}$. In autoISF, this is separately tuneable via: * higher ISFrange_weight * <i>iAPS</i>: “ISF weight for higher BG”, respectively via * lower ISFrange_weight * <i>iAPS</i>: “ISF weight for lower BG”	autoISF dynamicISF	see section 4.5.7 in: “FCL-e-book” https://github.com/bernie4375/FCL-potential-autoISF-research-

	Setting weights to zero provides no change vs profile_ISF		
bg source	the blood glucose source is the source where your bg values come from. They come from a CGM system which you wear through some kind of integration software like BYODA , xDrip+	CGM / FGM	Wiki - BG source
BMI	body mass index		https://www.diabettech.com/artificial-pancreas/losing-while-looping-can-you-lose-weight-with-a-diyaps/
Bolus wizard	See Calculator		
Boost	dev variant of AAPS involving simple Meal Announcement (MA) that can be stretched into a FCL		FCL-e-book 13.3.1; https://discord.gg/nYC4T9PgCR
brake	bg_brake_ISF can be separately tuned in autoISF at detected “braking”, i.e. when the bg rise slows down (at deceleration)	acceleration	see section 4.4 in: “ FCL-e-book ” https://github.com/bernie4375/FCL-potential-autoISF-research-
BYODA	Build Your Own Dexcom App - a special way to generate your own Dexcom App for reading out the transmitters and pass smoothened bg values on for looping (e.g. BYODA -> AAPS; or w/G6 also BYODA -> xDrip+ -> AAPS) while retaining the option to use Clarity® (as your doc office may want you to use)	xDrip+	Dexcom G6 ...ONE ... G7 https://docs.google.com/forms/d/e/1FAIpQLScD76G0Y-BIL4tZljaFkjlwughT83QIFM5v6ZEfO7gCU98iJQ/viewform
Calculator	HCL systems (and pump therapy in general) come with bolus calculators for suggesting bolus size for meals based on: to-be-digested g of carbs (while bolus is very active; later carbs -> eCarbs!); IC; ISF (if bg not near target); iob.	IC pre-bolus	https://androidaps.readthedocs.io/en/latest/Getting-Started/Screenshots.html#bolus-wizard https://loopkit.github.io/loopdocs/operation/features/bolus/#meal-bolus
calibration (of CGM)	if your symptoms disagree with what the CGM shows: test with your blood glucose meter; calibration is one (but not always the best) option then	CGM	https://navid200.github.io/xDrip/docs/Calibration.html
carb absorption	1) foods with slower absorption are easier to manage with insulin 2) 30 g/h seems a max (heavy eaters: do not bolus for more g than digested while your bolus goes strong!) 3) for oref systems, the lower border of plausibility is defined by the min5mCarbImpact 4) loops calculate delta cob from bg delta and iob delta (using carb ratio <u>and</u> ISF); more see at: dynamic carb absorption	cob; iob iob delta; dynamic carb abs. eCarbs; FPU; insulin kinetics	https://github.com/danamlewis/artificialpancreasbook/blob/master/8.-tips-and-tricks-for-real-life-with-an-aps.md#heres-the-detailed-explanation-of-what-we-learned https://loopkit.github.io/loopdocs/operation/features/carbs/#review-carb-absorption https://loopkit.github.io/loopdocs/operation/algorithm/prediction/?h=mar#dynamic-carbohydrate-absorption

	5) note that drugs, e.g. Ozempic® or comorbidities, e.g. gastroparesis have profound effects (inform yourself about implications re. carb absorpt. corridor).		https://androidaps.readthedocs.io/en/latest/Usage/FullClosedLoop.html#meal-related-limitations
carb ratio	see IC factor (AAPS) or carb insulin ratio CIR (iOS loops) for this	IC CIR	“Carb ratio...pdf” in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings
caregiver app	Android: -> AAPS client, iOS: -> Loop caregiver		
CGM	continuous glucose monitor (Dexcom, Libre, and other systems)	bg source G7, 6...; Libre 3	https://www.diabettech.com/cgm/six-of-the-best-digging-further-into-the-statistics/
CIR (carb insulin ratio)	factor (g/U) describing how many grams of carb are covered by one unit of insulin	carb absorption <i>AAPS</i> : IC	https://loopkit.github.io/loopdocs/operation/features/carbs/#review-carb-absorption
circadian (sensitivity, basal rate, ISF...)	basal need, carb ratio and ISF vary over 24 hours according to a „circadian“ pattern of varying sensitivity to insulin. Improper profile settings will “use up and waste” some of the loop system’s capability to correct for disturbances.	disturbance	see “ISF determ...pdf” : https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings ; chapter 5.
Closed Loop	closed-loop systems make automatic adjustments to basal delivery (TBR), without needing user-approval, based on an algorithm ; some also can automatically bolus (SMB)	Open Loop	https://androidaps.readthedocs.io/en/latest/Resources/clinician-guide-to-AndroidAPS.html#for-clinicians-a-general-introduction-and-guide-to-aaps Wiki closed loop
clinician support of DIY systems	the references given demonstrate increasing consensus to support DIY solutions as suitable for their patients		https://androidaps.readthedocs.io/en/latest/introduction.html#support-for-diy-looping-by-other-clinicians
cob (g)	carbs on board is the amount of carbohydrates currently available for digestion (“that still needs iob ”).	carb absorption; iob	https://androidaps.readthedocs.io/en/latest/Usage/COB-calculation.html#how-does-aaps-calculate-the-cob-value https://loopkit.github.io/loopdocs/operation/features/carbs/#review-carb-absorption
connectivity	numerous options for Bluetooth or WLAN connected devices. Additional open-source software and platforms (which are not shown in reference, e.g. Automate!, or Android Auto) can also be integrated.		https://androidaps.readthedocs.io/en/latest/introduction.html#what-is-the-connectivity-of-the-aaps-system
control of bg (sluggishness)	balancing carb absorption with insulin activity is a very difficult „sluggish“	carb absorption;	“Meal Mgt. 1 -4...pdf” in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings

	control problem - very much like boating. See slides in "Meal Mgt....pdf".	insulin kinetics; bg	
correction factor	scales size of autobolus (SMBs) in iOS loops - see delivery ratio		https://loopkit.github.io/loopdocs/operation/features/bolus/#recommended-bolus
customization <i>for iOS Loop</i>	advanced users of iOS Loop are offered several dev features <i>Other loops, or variants of them, have dev features integrated; and also offer middleware or Automations for the user to further customize</i>		https://www.loopandlearn.org/custom-code/
deceleration	the bg rise slows down (going towards peak)	brake	
delivery limit	max bolus and max basal rates allowed by your settings		
delivery ratio	portion of insulin required that gets delivered by the loop via SMB or TBR; fixed =50% in AAPS Master. Note that going towards 100% would be very unsafe at CGM jumps! It is better to wait another 5 minutes to confirm the bg trend, for getting again 50% of what <i>then</i> is required (=of what was held back, plus of "newly developed need") <i>Equivalent in iOS Loop: correction factor (=40%)</i> <i>Equivalent in Trio and iAPS: correction factor (<100%)</i>	insulin required SMB	https://androidaps.readthedocs.io/en/latest/Usage/FullClosedLoop.html#enabling-boosted-smbs-safety FCL-e-book section 2.3
delta All 3 delta categories show in the top section of the AAPS main screen	<u>delta</u> bg =d5=in past 5 minutes: important anchor point for loop calculations (see e.g.. in SMB tab of AAPS) = bg(0m)-bg(-5m) <u>short avg delta</u> = d15=avg. of <i>last</i> 3 deltas = (bg(0m)- bg(-15m))/3 <u>long avg.delta</u> =d45 is the average delta <i>between 15 and 45</i> minutes back = (bg(-15m)- bg(-45m))/6	iob delta	
desmos graph	see at sigmoid (dynamicISF calibr.help)		https://www.desmos.com/calculator/s9jxdmqhh8
dev	dev version of Master =software in pre-Master-release testing, dev variants have different, often extra, features to Master = still in development, insufficiently tested for broader release	autoISF; Boost; AIMI & others	
Dexcom	CGM , see G7, G6		
DIA (hours)	duration of insulin action	insulin kinetics	Wiki insulin types DIABETTECH - DIA "Insulin_DIA...pdf" in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings

disturbance	factors like meals and exercise (and ~40 others) disturb the smooth operation that would be possible with a well set basal profile. The set ISF , or a temp. activated exercise mode may enable the loop to automatically manage the disturbance. In other cases, a %profile switch or other measures may be needed.	ISF; exercise mode; %profile switch	https://diatribe.org/poster-now-available-42-factors-affect-blood-glucose FCL-e-book , section 5.2
DIY	see Open Source (that term describes better the cooperative and transparent efforts behind the “DIY” AID systems)		
DIY cockpit	term used for * having all buttons to “tweak” loop aggressiveness on the main screen of the closed loop phone * using tools like “ user action Automations ” in AAPS to construct extra buttons for this purpose These can be programmed to show only in pre-defined times, or geo-locations ...		FCL-e-book , section 5.2.2
Dual Hormone Loop	“Double closed loop” featuring insulin AND glucagon (in development): the glucagon component not only helps stay out of hypos. It enables a more aggressive treatment for preventing, or reducing, high glucose values, as well	glucagon	FCL-e-book , section 13.6
dura_ISF	...modulates profile_ISF at sluggish rise into a bg plateau, or at late/high bg plateaus; tuneable via the parameter duraISF_weight. <i>iAPS</i> : “duration weight”	autoISF	see section 4.5. in: “ FCL-e-book ” https://github.com/bernie4375/FCL-potential-autoISF-research-
duration weight	see dura_ISF	dura_ISF	
dynamic carb absorption	every 5 minutes, loops figure out carb absorption from bg delta , insulin activity consumed, and other data, and make predictions used in their dosing decision: 1) oreof loops (AAPS, Trio, and iAPS) can work entirely without carb inputs at meals (“UAM”, “FCL”) 2) iOS Loop makes strong use of the user’s carb inputs	UAM FCL	see “Carb ratio ...pdf” in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings/blob/HCL--settings-main-repo-(pdf)/chapter 1.2 1) https://androidaps.readthedocs.io/en/latest/Usage/COB-calculation.html 2) https://loopkit.github.io/loopdocs/operation/algorithm/prediction/ and https://loopkit.github.io/loopdocs/operation/algorithm/prediction/#carbohydrate-effect
dynamic carb ratio	automatic adaptation of IC to bg level and to past day(s) TDD (not useful in advanced oreof looping/FCL)	Trio, iAPS	https://discord.gg/gKXW5uX3m
dynamicISF	automatic adaptation of ISF to bg level and to past day(s) TDD; tuneable;		https://androidaps.readthedocs.io/en/latest/Usage/Dyna

	(Caution: Can make life easier but can be inferior to using a well tuned profile ISF + being proactive with manual or automated %profile switches) several refined versions exist, e.g.: * using bg AND predicted bg * pegging (in some form) to profileISF.	logarithmic dynISF sigmoid	micISF.html#dynamicisf-dynisf https://triadocs.org/usage/features/dynamic-isf/ see "04_ISF_weight_effects ...xls" in: https://github.com/bernie4375/FCL-potential-autoISF/blob/FCL-e-book/
dynamic iobTH	iob threshold above which no more SMBs are given in autoISF can automatically vary with a TT (a set temp. target): At elevated TT in exercise mode, a lowered iobTH lowers iob. At lowered TT ("EatingSoon"), iobTH can also be automatically elevated.	iobTH exercise	FCL-e-book section 6.1.3 (and 2.4, 2.6)
dynamic bg target	your loop probably allows you to generally select "sensitivity raises bg target" and "resistance lowers bg target". (Caution: Can lead to rollercoasters, especially if your carb settings and daily inputs are not spot-on (=> skewed Autosens!))		
EatingSoon TT (mg/dl) or (mmol/L)	Concept going back to looping pioneer Dana Lewis: to set a very low temp. bg target ~ 1 h before meals, so the loop gets a low bg starting point, and also some pos. iob at meal start	pre-bolus	https://github.com/danamlaw/artificialpancreasbook/blob/master/8.-tips-and-tricks-for-real-life-with-an-aps.md#how-to-do-eating-soon-mode
EatingNow	dev variant of AAPS involving simple Meal Announcement (MA) that might be stretched into a FCL		FCL-e-book 13.3.3; https://discord.gg/XqhnPRChEP
eCarbs	"extended carbs" = carb inputs split up over several hours; consider also effects from fat/protein (FPU) here. extended boluses you might know from regular pump therapy do not make much sense when looping; when your given bolus fades out, the loop takes increasingly over w/ SMBs or high %TBR	FPU SMB Calculator	https://androidaps.readthedocs.io/en/latest/Usage/Extended-Carbs.html#what-are-ecarbs-and-when-are-they-useful eCarbs use case
Emulator	program to analyze AAPS logfiles, including what-if analysis <i>Note: iAPS has some on-board analytic capabilities</i>	log files	https://github.com/autoisf/what-if
exercise mode (general)	a loop mode which limits how high iob will/can go, via any combination of: raising glucose target, lowering profile basal, elevating ISF, limiting iob.	TT %profile switch;	https://loopkit.github.io/loopdocs/operation/features/overrides/?h=exercise#create-an-override-preset

			https://androidaps.readthedocs.io/en/latest/Usage/making-sport-with-AAPS.html#cycling https://diyaps.org/.../how-to-exercise-when-exercise-is...
exercise mode (special features with HBET)	via a half-basal exercise target (HBET) in the settings, plus using any elevated temp. target (TT), basal rate, ISF and – in autoISF – also iobTH can <i>automatically</i> adjust for much milder insulin delivery	dynamic iobTH	https://github.com/bernie4375/FCL-potential-autoISF-research : -section 6.1.3.1 and xls subsheet “Exercise” in section “04_ISF..effects.xls”
extended bolus	frequently desired by looping beginners “to fight high bg”, this contradicts the very idea of looping: the algo must receive the inputs to manage bg (tuning). Boli (also the initial meal bolus in HCL) disturb the workings of the loop (that shuts off for a while via zero-temping)	eCarbs	https://androidaps.readthedocs.io/en/latest/Usage/Extended-Carbs.html#extended-bolus-and-why-they-won-t-work-in-closed-loop-environment
FCL-e-book	Series of pdfs about FCL , with case studies (autoISF focused, but all other methods are presented and referenced)	FCL	https://github.com/bernie4375/FCL-potential-autoISF-research -
FPU (g)	Fat-Protein-Units, converted into g carb equivalent Rather than worrying too much about conversion factors for FPU (controversy see slide 30 in 2nd link ->)oref loopers should rather see to it that their loop can deal well with temporary (!) insulin resistance from fatty acid receptor blockages (3rd link) Note that autoISF has the “dura_ISF” component to deal with plateaus of high bg.	eCarbs	https://iaps.readthedocs.io/en/latest/settings/services/fat-protein.html#fat-and-protein-conversion ; p. 3 in: “Meal Mgt. 3 .pdf” in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings https://androidaps.readthedocs.io/en/latest/Usage/Full-ClosedLoop.html#stagnation-at-high-bg-values
FCL: Full Closed Loop	Mode of closed looping <u>without</u> the user giving any boli, and without carb inputs. Depending on lifestyle and %TIR expectation, can run fully hands-off, or require a few button pushes at special disturbances , like heavier exercise. Setting up (personalized tuning) is difficult!	Hybrid Closed Loop (HCL) UAM	https://androidaps.readthedocs.io/en/latest/Usage/Full-ClosedLoop.html ; FCL-e-book see: https://github.com/bernie4375/FCL-potential-autoISF-research -
G7, G6, ONE, G5	abbreviation for Dexcom sensor/transmitter CGM systems	BYODA	Wiki - BG source
G6 x 2 (overlapping)	method to get un-interrupted CGM values	Anubis; xDrip Variant	FCL-e-book : Case study 1.5
git	git in our context here is the tool to mainly download the AAPS sources from Github for the build process. It's version-control system for tracking changes in computer files and coordinating work on those files especially for teams. -> necessary for apk updates		Wiki - update APK

GitHub	web-based hosting service for version control using git -> storage of source code to build apk, and of related documentation for: 1) AAPS 2a) Trio 2b) iAPS 3) iOS Loop <i>Note:</i> Special dev variants are on other Github pages, see e.g. @ autoISF	1.Android Studio; 2/3.Xcode	1. GitHub AndroidAPS; 2a.https://github.com/nightscout/Trio 2b.Github_build_iAPS.yml 3.https://loopkit.github.io/loopdocs/gh-actions/gh-first-time/
glucagon	... as experimental (!! adjunct for improved glucose control... Discord group “insulin-plus-glucagon”: https://discord.gg/eHSgx5jWuk	Dual Hormone Loop	https://www.diabettech.com/glucagon/glucagon-me-n1-experiments-with-microbolusing-glucagon-part-2-quantifying-use/
glucose momentum	term used for predictions in iOS Loop; assumes that the bg effects seen in the last three five-minute segments are likely to continue for a short period of time	iOS Loop	
glucose target	corrections by the loop aim at the bg target value (set in the profile* for each hour of the day); depending on nature of disturbances, and properly set ISF, that value should be gradually reached over the course of 2-4 hours; *note that targets might get auto-adjusted (e.g. by Autosens, or in an exercise mode).	TT	
half-basal exercise target	at <i>elevated</i> temp. glucose targets (as for exercise), loop aggressiveness (profile basal, ISF used) gets reduced. Effect is the stronger the lower this parameter is set (in AAPS/Preferences default is 160 mg/dl*), and the higher the TT; in case you also switch “<i>low_TT_reduces_sensitivity</i>” ON in /Preferences, the parameter can *) also be used for <i>increasing</i> the loop aggressiveness!	* AAPS x autoISF; might differ or be non-existent in other loops	Details see on Exercise Mode page of the autoISF Quick guide: https://github.com/gazelle/autoISF/blob/A3.2.0.4_ai3.0.1/autoISF3.0.1_Quick_Guide.pdf or newer
HCL: Hybrid Closed Loop	the usual mode of looping, with the user initiating a meal bolus (and making other frequent inputs, notably re. carbs). This is really a compromise owed to slow insulins in-capable of dealing with rapid carb absorption	calculator; extended bolus; FCL	https://androidaps.readthedocs.io/en/latest/introduction.html#what-does-hybrid-closed-loop-mean
iAPS	oref loop (like AAPS, but) for i-phone Caution: iAPS is a rapidly evolving DIY loop variant with a weekly stream of new features and bugfixes, limited documentation, and a need to constantly stay informed in Discord. <i>Beginners (on “oref-“ loops) may be safer to start on Trio.</i> Source code (apk): see GitHub Requires Apple developer licence (\$ 100/year), and Xcode.	Github Trio AAPS Xcode	https://iaps.readthedocs.io/en/latest/ https://discord.gg/JVXwG7gS https://www.facebook.com/groups/1351938092206709

IC (carb ratio) (g/U)	factor (g/U) used in AAPS to describe how many grams of carb are covered by one unit of insulin	<i>iOS Loop:</i> CIR	“Carb ratio...pdf” in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings
ICE (insulin counteraction effects)	example: user enters 72g carbs w/ 4 h absorption => <i>Loop</i> adds 50% time, and assumes a min. absorption rate MAR of 12g/hr for 6 hrs. This linear model is modulated using recently observed glucose data to estimate how fast carbohydrates have been absorbing. The expected change in glucose due to insulin effects alone is compared to the actual observed changes in glucose. This difference is termed the insulin counteraction effect (ICE):		https://loopkit.github.io/loopdocs/loop-3/displays-v3/?h=ice#ice-chart
individualized tuning	DIY loops are not self-learning but require “tuning” to find proper <u>individual settings</u>. <u>1) for Meal Management</u> HCL: AAPS Objectives; meal management FCL: dial in your settings (incl. Automations) so the loop is enabled to mimick your successful HCL Meal Management (notably, similar insulin activity curve, going up a bit later, but very steep....) <u>2) finding individual temporary settings to adapt loop aggressiveness for other disturbances</u>, e.g. exercise Note 1: Tuning must follow a certain sequence (to avoid instability from counter-balanced multiple errors). Resist the temptation to just play around on the many “buttons” offered! Do a good job with “vanilla” Master, first. Note 2: Learn not to interfere, make your loop – over time – fit to manage automatically	Objectives; Meal management; FCL tuning disturbance exercise vanilla	HCL guidance in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings https://loopkit.github.io/loopdocs/loop-3/therapy-settings/ FCL-e-book in: https://github.com/bernie4375/FCL-potential-autoISF-research- 2) sections 5 and 6 in FCL-e-book: https://github.com/bernie4375/FCL-potential-autoISF-research-
insulin activity (U/5 min)	part of iob that will become active in the upcoming 5 minutes (above profile basal supply => figure can be negative also)	insulin kinetics	
insulin counteraction effect (ICE)	ICE describes deviations when the observed changes in glucose do not quite match up with the expected change due to insulin effects alone -> $\text{absorbed_carbs} = \text{ICE} * \text{CIR} / \text{ISF}$	iOS Loop bg deviation	https://loopkit.github.io/loopdocs/loop-3/displays-v3/?h=ice#ice-chart
insulin kinetics	AAPS insulin tab shows two curves: The <u>pink</u> curve starts at 1.0 (100%) and goes down to 0 (0%) when the DIA is	control of bg (sluggishness)	“Insulin_DIA...pdf” in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings

	over. It shows iob left, at any time. The <u>blue curve</u> shows how the activity goes: Practically nothing (!) for a bunch of minutes, then rapidly going high, and then slowly fading out over the DIA period (with a maximum at time-to-peak). For its calculations, AAPS adds these blue curves up for all boli, SMBs and TBRs profile basal -> <u>thin yellow</u> “activity” curve you can see in your AAPS glucose screen!		“The artificial pancreas...pdf” in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings/blob/FCL-w/autoISF/ https://loopkit.github.io/loopdocs/operation/algorithm/prediction/#insulin-effect
insulin required (U)	<u>key parameter in the <i>oref</i> loop algo:</u> from how bg, iob and cob resp. carb deviations develop (-> predictions), need for more insulin is calculated <u>equivalent (but determined differently) in <i>iOS Loop</i>:</u> -> recommended dose	SMB predictions delivery ratio	https://openaps.readthedocs.io/en/latest/docs/While%20You%20Wait%20For%20Gear/Understand-determine-basal.html?highlight=insulin%20required#blending-relevant-predictions
integral correction effect	via selecting the IRC setting for “Integral ..Correction”, <i>iOS Loop</i> uses “learning” from historical prediction problems (deviations) when making glucose predictions	iOS Loop	https://loopkit.github.io/loopdocs/operation/algorithm/prediction/#integral-retrospective-correction-effect
iob (U)	insulin on board; units of insulin (above basal need) <u>currently available to become (within the remainder of its DIA) active</u> in your body (to deal with un-absorbed carbs, or with other disturbances)	insulin activity DIA	https://androidaps.readthedocs.io/en/latest/Getting-Started/Screenshots.html#section-d-iob-cob-br-and-as
iob delta (U)	insulin consumed = (1) delta bg / ISF = used for bg correction (2) the rest of the delta iob, multiplied with IC, is the grams of carbs absorbed. (3) if (2) results in <u>implausible carb absorption</u>, then IC and ISF are adapted “to force a plausible fit”; and the adapted insulin sensitivity is then reflected in <u>Autosens ≠ 100%</u>	carb abs. 2), 3) min5m_carb.impact	
iobTH (U) or iobTH% (% of maxIOB)	iob threshold (set below maxIOB); at iob > iobTH, the loop will give no more boli (SMB) but only TBR	iob; maxIOB; SMB	https://androidaps.readthedocs.io/de/latest/Usage/FullClosedLoop.html#iob-threshold
iOS Loop	easy DIY loop to set up on i-phone; <i>iOS Loop</i> uses a model predictive control (MPC) algorithm to maintain glucose in a correction range by predicting the contributions from four individual effects (insulin, carbohydrates, retrospective correction, and glucose momentum).	i-phone loops with different algorithm (as in AAPS) =>: Trio; or iAPS	https://loopkit.github.io/loopdocs/ https://www.loopandlearn.org/starting-loop/ https://loopkit.github.io/loopdocs/fags/algorithm-fags/#more-algorithm-information

	algorithm requires precise carb inputs at all meals (<u>no</u> UAM or FCL); very limited choices of pumps customization options		https://loopkit.github.io/loopdocs/operation/algorithm/overview/#algorithm-terminology
IRC	see integral correction	iOS Loop	
ISF (mg/dl)/U or (mmol/L)/U	insulin sensitivity factor = the expected decrease in bg as a result of one unit of insulin; most important parameter in oref loops	IC	“ISF determ...pdf” in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings
ISF_weight	factor used to tune components of autoISF	autoISF acce_ISF bg_ISF dura_ISF pp_ISF	“04_ISF_weight_effects ..xls” in: https://github.com/bernie4375/FCL-potential-autoISF/blob/FCL-e-book/
LGS	Low Glucose Suspend = AAPS will reduce basal if bg is dropping. But if bg is rising then it will only increase basal if the iob is negative (from a previous LGS), otherwise basal rates will remain the same as your selected profile. You may temporarily experience spikes following treated hypos without the ability to increase basal on the rebound.	<i>AAPS:</i> objective 6	https://wiki.aaps.app/en/latest/Usage/completing-the-objectives.html#objective-6-starting-to-close-the-loop-with-low-glucose-suspend
Libre 3	CGM ; also Libre2 (alternatives to Dexcom CGMs)	CGM bg	https://www.diabettech.com/cgm/battle-royale-freestyle-libre-3-and-dexcom-g7-face-off-the-results/ ;
Libre 3 1 minute	first option to run a 1-minute CGM - which could bridge a few minutes of “sluggish” delay in looping. This is particularly of interest in no-bolussing FCL (see 2 nd reference ->).	bg; control (sluggishness)	https://github.com/Nightsco.../xDrip/releases/tag/2023.02.15 https://github.com/gazelle/autoISF
log files	record of all AAPS actions (useful for troubleshooting and debugging)		Wiki - log files
logarithmic dynamicISF	default variant of dynamicISF, with a logarithmic strengthening of applied ISF with rising glucose values; chart is depicted here: https://www.desmos.com/calculator/zrkugmdnob?lang=de	dynamic ISF sigmoid	https://www.facebook.com/groups/diytrio/permalink/1488923685174815/ “04_ISF_weight_effects ..xls” in: https://github.com/bernie4375/FCL-potential-autoISF/blob/FCL-e-book/
Loop	see iOS Loop		
Loop caregiver app	App that enables remote control (e.g. kid/parent) in iPhone based DIY loops	<i>Android equivalent:</i> AAPS Client	https://loopkit.github.io/loopdocs/nightscoout/loop-caregiver/

Loop follow	App that enables remote data viewing (e.g. kid/parent) in iPhone based DIY loops	AAPS Client	https://www.loopandlearn.org/loop-follow/
low glucose suspend	see LGS	LGS	
MA	see: Meal Announcement		
MAR (minimum absorption rate)	example: user enters 72g carbs w/ 4 h absorption => <i>iOS Loop</i> adds 50% time, and assumes a min. absorption rate (MAR) of 12g/hr for 6 hrs.	<i>oref</i> <i>equivalent:</i> min_5m_carb-impact	https://loopkit.github.io/loopdocs/operation/algorithm/predictions/?h=mar#dynamic-carbohydrate-absorption
Master	Master is the latest official release, the software that should be used. Note that it is advisable to tune profile in Master before adding more features.	dev; vanilla	
maxIOB	safety feature: maximum total iob the loop can't go over. (set by the user; but can be hard-limited by set patient type!)		https://androidaps.readthedocs.io/en/latest/Usage/Open-APS-features.html#maximum-total-iob-openaps-cant-go-over-openaps-max-iob
MDI	multiple daily injections: option to manage your t1d with an insulin pen (and bg measurements or CGM). An option you should resort to in case components of your loop system are unreliable (pump, occlusion , erratic CGM, instable Bluetooth)		https://androidaps.readthedocs.io/en/latest/introduction.html#how-does-aaps-compare-to-mdi-and-open-looping
Meal Announcement (MA)	MA is a closed looping mode between HCL and FCL : In contrast to HCL, no carbs are counted with an attempt to give a suitable meal bolus. But in contrast to FCL, some form of meal announcement must be made, usually by giving a small pre-bolus . PS: Another variant of MA could be to announce carbs, but not to bolus. That makes no sense with <i>oref</i> loops.		sections 7. and 13.3 in: https://github.com/bernie4375/FCL-potential-autoISF/blob/FCL-e-book/
Meal Management	Juggling (for every meal!) the differing carb and insulin absorption characteristics, so bg stays in range, is a tough, if at all possible, mission. Big effort should go into individualized tuning of the loop system, and into defining bolus strategies	EatingSoonTT; pre-bolus	„Meal Mgt.1-4.pdf“ and „Carb ratio ...pdf“ in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings
middleware	custom algorithm add-ons (notably in iAPS that does <u>not</u> have the Automation feature of AAPS) “Middleware gives you every algo setting to tweak. You just need a couple of TT presets that you activate directly or via Shortcut with other stuff and whatnot.- Middleware just looks for the specific target value, and tweaks whatever parameter set you want changed (Not as	<i>AAPS</i> <i>equivalent:</i> Auto-mations	https://github.com/macconnellk/RoboSurfer/tree/main Middleware code for iAPS https://discord.gg/3JWQRzfyB2

	nice as activating an override, but easily possible)’” (Robert, Discord June 2025).		
min_5m_carb impact	safety feature (oref): default carb decay at times when dynamic carb absorption does not reasonably work out based on your bg reactions	carb absorption <i>iOS Loop:</i> <i>see</i> MAR	„min5m_CI...xls” in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings Wiki - config builder
minimum carb absorption rate	<i>see for oref:</i> min_5m_carb-impact <i>see for iOS Loop:</i> MAR		
negative iob	iob is defined as insulin on board above (profile) basal need. Negative iob can occur (and can self-resolve, too). Too high set <i>profile</i> basal can be behind neg.iob. Likewise, if you forget to keep <i>temp.%profile</i> reduced after a day of exercise, your profile basal will be <i>temporarily</i> too high, and neg.iob would be reported. Tipp: For easy spotting of neg.iob phases you can use an extra graph on the bottom of your AAPS main screen; define IOB (not ABS!) as the first parameter		https://androidaps.readthedocs.io/en/latest/SettingUpAaps/YourAapsProfile.html#i-have-negative-iob-is-this-an-issue https://loopkit.github.io/loopdocs/faqs/algorithm-faqs/?h=negative+iob#what-does-negative-active-insulin-mean
Nightscout (NS)	open source project to access and report CGM and related data. Convenient options (10be, NS pro) that host your large amount of data come with monthly cost. - Except for a brief initial period, it is not required that AAPS loopers are using NS.	Nightscout Reporter	Nightscout Hosted services: https://nightscout.github.io/#nightscout-as-a-service
Nightscout Reporter	free tool provided by a fellow looper to generate great PDF reports from your Nightscout data, e.g. for meetings with your diabetes team.	Nightscout	Nightscout Reporter NS Reporter @ Facebook
NS Client	part of AAPS to connect to your Nightscout site; important also for remote monitoring and control (parent/kid)	remote control	Wiki - NS Client
Objectives	learning program within AAPS guiding you step by step from open to closed loop		Wiki - objectives
occlusion	insulin the pump releases is not fully delivered in the body => persistent very high bg despite (fake) high iob – dangerous, must be avoided! ((Often occlusions are only partial, and then dynamicISF might help a bit. But try to avoid!))		“Occlusion..pdf” in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings
OpenAPS	open artificial pancreas system: runs on small computers (i.e. Raspberry Pi) AAPS and Trio/iAPS use most of the OpenAPS features		OpenAPS docs

Open Loop	system will only <u>suggest</u> adjustments which have to be confirmed manually in the application	Closed Loop	Wiki - config builder https://loopkit.github.io/loopdocs/operation/loop/open-loop/
Open Source	philosophy to openly share product (especially, software) development without profit orientation, and not operating in narrow frameworks like mandated by e.g. regulations on medical products, (Alternatively, the prefix “DIY” is often used)	Github clinician support	https://androidaps.readthedocs.io/en/latest/Getting-Started/Introduction.html#no-commercial-constraints-due-to-open-application-interfaces
oref	the key algorithm behind OpenAPS, AAPS and iAPS. In SMB+UAM setting, it enables looping without any carb inputs	dynamic carb absorption iOS Loop;	Wiki - sensitivity detection https://openaps.readthedocs.io/en/latest/docs/While%20You%20Wait%20For%20Gear/Understand-determine-basal.html#understanding-the-basic-logic-written-version
override	see aggressiveness of the loop <i>equivalent in oref:</i> -> profile switch		https://loopkit.github.io/loopdocs/operation/features/overrides/?h=override https://loopkit.github.io/loopdocs/operation/features/overrides/?h=exercise#create-an-override-preset
parabola fit	mathematical method used in autoISF for 1) improved smoothing and predicting of bg 2) early detection of -> acceleration	autoISF acceleration smoothing	See last page before attachments in “Guide to VDF Files for the AAPS Emulator.pdf” at /Documentation in English of: https://github.com/gazelle/APS-what-if/
peak time or time-to-peak (minutes)	time to maximum effect of insulin given: shorter is better for looping, but also exposes bad tuning and can be unsafe (hypos!) for looping beginners	insulin kinetics	Wiki - config builder “Insulin_DIA...pdf” in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings
pp_ISF	... modulates ISF more aggressive when high deltas are seen; this is tuned via pp_ISF_weight <i>iAPS: “ISF weight for post-prandial bg rise”</i>	autoISF	see section 4.3. in: “FCL-e-book” https://github.com/bernie4375/FCL-potential-autoISF-research-
pre-bolus	any meal containing rapid carbs will push bg high faster than insulin could become strongly active (nearing peak-time) to control this. Hence giving your meal bolus a number of minutes <i>before</i> meal start can be a good idea.	Meal Mgt. Calculator Eating SoonTT	„Meal Management 2 .pdf” in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings https://loopkit.github.io/loopdocs/operation/features/carbs/#time
predictions	loops look into predictions (bg in the future), not just on present bg , for their decisions	insulinReq u.	1). For AAPS see Wiki - prediction lines

	1).In oreb systems, this is based on several different calculations; eventualBG uses traditional bolus calculator math. IOBpredBGs predicts only an eventual BG value, once all insulin activity takes effect. ZTpredBGs what will happen in the “worst likely case,” if observed carb absorption suddenly ceases, and a zero-temp is applied until BG begins rising at/above target. COBpredBGs is calculated based on observed deviations since carb entry, assuming that carbs would continue to be digested/absorbed at a configurable minimum rate. UAMpredBGs Once deviations have peaked, UAM calculations assume that the deviations will continue decreasing at that same rate until they reach zero. If they’re decreasing, but too slowly, it assumes they’ll decrease linearly to zero over 3 hours 2).for iOS Loop, see @ retrospective correction, and @ iOS Loop		https://openaps.readthedocs.io/en/latest/docs/While%20You%20Wait%20For%20Gear/Understand-determine-basal.html#understanding-the-purple-prediction-lines 2) for iOS Loop’s (one) prediction see: https://loopkit.github.io/loopdocs/operation/algorithm/prediction/#glucose-prediction
profile	basic treatment settings (basal rate, DIA, carb ratio, ISF, bg target) AAPS v3 only supports local profiles but Nightscout profiles can be copied (synchronized) to AAPS *note that some profile settings might get temporarily auto-adjusted (e.g. by Autosens, or in an exercise mode).		Wiki - profile
profile switch (% other than 100)	temporary (= assigned with a duration) change of profile used <u>reflecting percentual increase/decrease of insulin sensitivity</u> (e.g. <<100% in an exercise context) <i>equivalent in iOS Loop: -> override</i>	aggressive ness of the loop	https://androidaps.readthedocs.io/en/latest/Usage/Profiles.html#profile-switch
readthedocs	documentations available for each DIY looping system	wiki	Links see at: AAPS, iOS Loop, iAPS, Trio
recommended dose	with each cycle, iOS Loop generates a glucose prediction and a recommended	correction factor;	https://loopkit.github.io/loopdocs/faqs/safety-

	dose (positive or negative) to bring you to your correction range <i>equivalent (but determined differently) in oref loops: -> insulin required</i>	delivery ratio	faqs/?h=dosing#understand-delivery-limits https://loopkit.github.io/loopdocs/operation/algorithm/auto-adjust/#calculated-dose
remote control	DIY looping systems come with options for parents/caregivers to remotely control their young kids' loops, e.g. via secure SMS commands or NS Client	NS Client AAPS Client Loop Caregiver	https://androidaps.readthedocs.io/en/latest/introduction.html#remote-control https://loopkit.github.io/loopdocs/nightscout/remote-overview/
retrospective correction effect	Going forward in 5 minute steps, the loop keeps observing deviations between predicted and actually seen glucose values, and translates this into a so-called Retrospective Correction Effect . Assuming these effects (from different temp. insulin sensitivity than suggested by the factors, or from wrong carb inputs) will continue for some short time, the loop can improve its prediction where bg is headed, and whether insulin other-than- profile basal need is required.	iOS Loop	https://loopkit.github.io/loopdocs/operation/algorithm/prediction/#retrospective-correction-effect
resistance	above-normal insulin need (e.g. reduced sensitivity to insulin after a fatty meal)	FPU sensitivity	
roller coaster	term to describe bg curves that go steep down, then up, then down again ... often a result of too aggressive ISF ; dynamic settings (ISF, bg target etc) can also increase the tendency towards r.c.	ISF; dynamic ISF ...	
sensitivity	high insulin sensitivity => below-normal insulin need (e.g. after exercise, that makes you temp. more insulin sensitive)	resistance exercise	
sensitivity adaptation	rather than invariably using average sensitivity data as set in the profile, <i>loops may be capable of</i> : 1) ->sensitivity detection -> Autosens 2) inferring sensitivity change based on bg or TDD -> dynamic ISF 3) inferring sensitivity from characteristic personal data patterns (-> Automation, middleware, customization) 4) inferring sensitivity from stepcount (-> Activity Monitor) 5) inferring sensitivity from bg curve characteristics (-> autoISF). 6) <i>the user</i> can make <i>manual</i> temp. adaptations, as well (e.g. set temp. target, temp.%profile)	aggressiveness	(links see under highlighted terms)
sensitivity detection	calculation of sensitivity to insulin (based on deviations that cannot be “explained”	Autosens	DIABETTECH - Autosens

	by carb absorption) . often a result of exercise, hormonal changes, etc.		
sensor noise	unstable CGM readings leading to "jumpy" values - <i>a big problem when using elevated delivery rates (correction factors), or "acceleration"-based boosts for FCL</i>	CGM smoothing	Wiki - sensor noise
sigmoid	uses profile ISF and adjusts it "in S-curve shape" with glucose level above target, and TDD ; can turn out more aggressive than standard dynamicISF if Autosens min/max is set wide open => <i>not recommended for Trio or iAPS beginners</i>	dynamic ISF logarithmic dynamicISF	https://www.desmos.com/calculator/s9jxdmqhh8 "04_ISF_weight_effects...xls" in: https://github.com/bernie4375/FCL-potential-autoISF/blob/FCL-e-book/
SMB	small bolus given by the loop (advanced feature for faster bg adjustment vs TBR); max size restricted via setting minutes of profile basal (30 .. 120). Note that small size is a precaution for beginners against wrong ISF or bad CGM. Try to "open up" when tuning; SMBs try to give 50% of -> insulin_requ. (see also -> (SMB) delivery ratio); zero basal right after a SMB is expected.	TBR <i>iOS Loop =</i> Auto()Bolus UAM iobTH	Wiki - SMB Wiki – AMA to SMB
SMB delivery ratio	defines which % (default 50 or 60%) of the calculated insulinRequ. shall be given now vs. waiting 5 more minutes, (and then again same % of what then is open, which includes the portion that had to wait). Caution: Using >75% not recommended as it does not provide room for CGM jitter, and reduces flexibility around SMB/TBR sizing to pull back on insulin delivery when required.	<i>iOS loops:</i> correction factor	FCL-e-book section 2.3
SMB range extention	Bolus sizes the loop can give are severely restricted in HCL (usually to max 2x hourly basal). This factor multiplies to magnify "allowed" SMB size in FCL .	<i>iOS Loop:</i> autoBolus max	FCL-e-book section 2.1
smoothing	CGM systems deliver raw bg values that can be too "jumpy" to use. The loop system and/or intermediate app that captures the transmitter signals (1) offer options to smooth the values into a "realistic" bg curve and (2) might (!) contain internal plausibility checks. Smoother is safer (may be needed), but it slows the loop's treatment of bg rises	CGM parabola fit	https://androidaps.readthedocs.io/en/latest/Usage/Smoothing-Blood-Glucose-Data.html#smoothing-blood-glucose-data https://www.diabettech.com/cgm/back-smoothing-or-not-back-smoothing-is-that-the-question/
source code	describes how the loop works in all details	apk Github	

	DIY loops are Open Source = free access on Github = anyone can read, use and branch out/change code (-> dev variants)		
TAI	Trio+autoISF	Trio	https://github.com/mountrcg/Tai
TBR (% of profile basal)	temporary basal rate (given as % of profile basal). Note that <i>elevated</i> TBRs regulate bg far slower <i>down</i> than SMBs .	SMB	https://loopkit.github.io/loopdocs/operation/algorithm/auto-adjust/#determine-the-temporary-basal-rate
TDD (U)	total daily insulin dose (bolus + basal per day); very roughly characterizes a person's insulin sensitivity Note that occlusions can produce very noticeable false high TDD values!	dynamic ISF; occlusion	
TIR (%)	% of time bg is in a 70 – 180 mg/dl (3.9 – 10 mmol/L) range. <i>All data must be reported (also) in this format, because it is close to med. recommendations and data cannot be re-calculated into other formats without access to all individual data points!</i>		https://diatribe.org/diabetes-management/time-range https://my.clevelandclinic.org/health/articles/time-in-range-diabetes
TITR (%)	% of time bg is in a TIGHT 70 – 140 mg/dl (3.9 – 7.8 mmol/L) range.		https://pubmed.ncbi.nlm.nih.gov/38787436/
Trio	oref loop like AAPS but for i-phone -> Trio (or iAPS). Source code (apk): see Github Building requires Apple developer licence (\$ 100/year), and Xcode.	Github Xcode iAPS AAPS	https://tridocs.org/configuration/new-user-setup/ https://open-iaps-docs.readthedocs.io/en/latest/ https://discord.gg/Rr37aAzWz9 https://www.facebook.com/groups/diytrio
Tsunami	dev variant of AAPS involving simple Meal Announcement (MA) that might be stretched into a FCL		FCL-e-book 13.3.4; https://discord.gg/veRKcgwVUT
TT (mg/dl) or (mmol/L)	temporary target: temporary increase /decrease of bg target (range) a) by the user, e.g. for exercise ; or for “ eating soon ” b) automatically by the loop, e.g. at detected sensitivity settings can allow to change bg target; or a user-defined Automation can set a TT => purpose: the loop will deliver a bit less / a bit more insulin by “shooting for” different targets. <i>Some loops offer an option to boost this further, see under “half-basal.”</i>	half-basal exercise target	Wiki - temp targets
TT (or target) even / odd	some loops offer to set different behaviors (SMBs allowed /blocked), with setting even/odd numbered TT (or also profile target)	autoISF SMB	

tuning	see: individualized tuning	vanilla	
UAM	Un-Announced Meals - Detection of significant increase in bg levels due to meals (but also adrenaline or other influences), and attempt to adjust this with SMBs. Carb inputs are optional in theoref loops (AAPS, Trio, iAPS)	dyn.carb absorption SMB FCL	Wiki - SMB Why no carb inputs needed see chapter 1.2.2 in „IC (carb ratio)..pdf” in: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings
UTZ, CET	time zones: The AAPS loop data are generally recorded in UTZ time (universal Greenwich time). Your AAPS screen will show your Smartphone time zone, like central European daylight saving time (CET DST).	logfiles	Wiki DST
vanilla	term often used for Master version. Advice is not to make use of extras (“bells and whistles”) before the basics are tuned in right. Reason: Errors can be balanced with counter-errors => instable system (...that is also hard to calibrate better later, and may require a complete new start)	Master	
virtual pump	option to try loop functions without a pump connected (=> manual enacting suggestions, <i>in AAPS up to Objective 5</i>)	Open Loop	https://androidaps.readthedocs.io/en/latest/SettingUpAaps/CompletingTheObjectives.html#objective-5-understanding-your-open-loop-including-its-temp-basal-recommendations
_weight (-) e.g. bgAccel_ISF_weight	tuning factors used in autoISF to adapt ISF according to developing glucose curve	autoISF	FCL-e-book , section 4
wiki	readthedocs (documentations, one for each DIY looping system)		<i>Links see at:</i> AAPS, iOS Loop, iAPS, Trio
Xcode	developer software (free, but \$ 100/y developer licence) needed to complete and maintain your personal copy of iAPS (or iOS Loop)	<i>for AAPS:</i> Android Studio	https://loopkit.github.io/loopdocs/gh-actions/gh-first-time/
xDrip+	open source software to read CGM transmitters and pass (if desired, smoothened) values on for looping	BYODA	https://navid200.github.io/xDrip/docs/FAQ_page.html https://jamorham.github.io/#xDrip-plus https://navid200.github.io/xDrip/docs/Installation_page.html
xDrip Variant	enables up to 4 parallel xDrip instances on smartphone	G6 x 2 (overlapping)	https://navid200.github.io/xDrip/docs/Variants.html FCL-e-book: Case studies 1.5 and 5.3
YDMV (your diabetes may vary)	sometimes used to remind readers that their body might react differently than		“42 factors...pdf” at: https://github.com/bernie4375/

	expected based on what they see reported by others		HCL-Meal-Mgt.-ISF-and-IC-settings
zero-temp (ing)	temporary basal rate with 0% (no basal insulin delivery); often seen after a bolus was given: Moving some basal (from baseline need, as defined in profile) <i>into the bolus</i> or <i>SMB</i> provides for fastest correction; in turn, basal supply is reduced until safe to continue	TBR	https://androidaps.readthedocs.io/en/latest/UsefulLinks/ClinicianGuideToAaps.html#scenario-1-zero-temp-for-safety https://loopkit.github.io/loopdocs/operation/algorithm/auto-adjust/?h=zero+temp#zero-the-basal-rate